LUP-########## ######## A # ############## #####, ######## ######## $Ax = b$, ########## ############## # ######### ########.

```
import numpy as np
import pandas as pd
from IPython.display import display

from mtx_utils import load_mtx

A, b = load_mtx("tasks/1_1.txt")
n = len(b)

A, b
```

```
(array([[ 1., -5., -7.,  1.],
        [ 1., -3., -9., -4.],
        [-2.,  4.,  2.,  1.],
        [-9.,  9.,  5.,  3.]]),
 array([-75., -41.,  18.,  29.]))
```

$$Ax = b,$$
$$P^{-1}LU = A.$$

```
# Permutations для строк в матрице
P = np.eye(n, dtype=float)

L = np.eye(n, dtype=float)
U = A.copy()
swap_count = 0

for col in range(n):
    pivot_row = col + np.argmax(np.abs(U[col:, col]))

    if np.isclose(U[pivot_row, col], 0.0):
        raise ValueError("Матрица вырожденная: LU-разложение невозможно")

    if pivot_row != col:
        swap_count += 1
        U[[col, pivot_row], :] = U[[pivot_row, col], :]
        P[[col, pivot_row], :] = P[[pivot_row, col], :]

        L[[col, pivot_row], :col] = L[[pivot_row, col], :col]

    for row in range(col + 1, n):
        factor = U[row, col] / U[col, col]
        L[row, col] = factor
        U[row, col:] -= factor * U[col, col:]

display(pd.DataFrame(U), pd.DataFrame(L))
```

	0	1	2	3
0	-9.0	9.0	5.000000	3.000000
1	0.0	-4.0	-6.444444	1.333333
2	0.0	0.0	-5.222222	-4.333333
3	0.0	0.0	0.000000	2.936170

	0	1	2	3
0	1.000000	0.0	0.000000	0.0
1	-0.111111	1.0	0.000000	0.0
2	-0.111111	0.5	1.000000	0.0
3	0.222222	-0.5	0.446809	1.0

$$Ux = z,$$
$$Lz = Pb.$$

```
from IPython.display import Math

pb = P @ b

z = np.zeros(n, dtype=float)
for row in range(n):
    z[row] = pb[row] - np.dot(L[row, :row], z[:row])

x = np.zeros(n, dtype=float)
for row in range(n - 1, -1, -1):
    s = np.dot(U[row, row + 1:], x[row + 1:])
    x[row] = (z[row] - s) / U[row, row]

for i in range(n):
    display(Math(fr"x_{i+1} = {x[i]:.3f}"))
```

$$x_1 = 2.000$$

$$x_2 = 4.000$$

$$x_3 = 7.000$$

$$x_4 = -8.000$$

```
detA = (-1) ** swap_count * np.prod(np.diag(U))
display(Math(fr"det A = {detA:.3f}"))
```

$$detA = 552.000$$

$$\begin{aligned} PA &= LU, \\ PAA^{-1} &= LUA^{-1}, \\ P &= LUA^{-1}. \end{aligned}$$

#########

$$\begin{aligned} C &= UA^{-1}, \\ LC &= P. \end{aligned}$$

```
A_inv = np.zeros((n, n), dtype=np.float64)

C = np.zeros((n, n), dtype=np.float64)
for row in range(n):
    C[row, :] = P[row, :] - L[row, :row] @ C[:row, :]

for row in range(n - 1, -1, -1):
    A_inv[row, :] = (C[row, :] - U[row, row + 1:] @ A_inv[row + 1:, :]) / U[row, row]

pd.DataFrame(A_inv)
```

	0	1	2	3
0	0.028986	-0.032609	0.525362	-0.228261
1	0.007246	0.054348	0.568841	-0.119565
2	-0.108696	-0.065217	-0.282609	0.043478
3	0.246377	-0.152174	0.340580	-0.065217

######## #######

```
np.allclose(x, np.linalg.solve(A, b))
```

```
True
```

######## ############ ###### L U # ############ #####

```
np.allclose(A, P.T @ L @ U)
```

```
True
```

######## ############ ############ ############

```
np.isclose(np.linalg.det(A), detA)
```

```
np.True_
```

######## ############ ######## #######

```
np.allclose(np.linalg.inv(A), A_inv)
```

```
True
```